Startup Application Form

English

2021秋季创业营

COMPANY

Company name
Ambient Systems 天眼查 烯牛数据

Company URL, if any
https://www.ambientsystems.ai/

Product name
Ambient Systems AI

If you have a demo, what's the URL? For non-software, demo can be a video.
https://youtu.be/wgknMOqMJo4

Describe your product in 50 characters or less.
AI solutions for smart buildings

What is your company going to make?
Ambient builds artificial intelligence solutions for cities of tomorrow. Our launch market is indoor farms, where we provide climate optimization to improve crop growth, reduce energy consumption, and improve worker productivity. We have launched the MVP and completed a pilot trial with a leading indoor farm in Singapore, demonstrating our patented technology.

What categories best apply to your company?
B2B, Robotics, Artificial Intelligence, Energy, Agriculture, Healthcare, Hardware, Sensors

Where do you live now, and where would the company be based after bootcamp?
Singapore / Singapore

How many founders are there with at least 5% equity in the company?
1

Please enter the url of an 1-minute video uploaded to Youku or other video platforms (please provide the password if it's private) introducing the founder(s). This video is an important part of the application.(Follow the Video Guidelines: www.miracleplus.com/videoguideline)
https://youtu.be/hbZOcqf4CnM

Video password (if any)

How long have the founders known one another and how did you meet? Have any of the founders not met in person?
There are a few other team members who joined recently. Everyone is based in Singapore and has met in person.

FOUNDERS

1st

Name
Ivan Damnjanovic

Email
ivan@ambientsystems.ai

Phone
65[redacted]

WeChat ID

Toronto, Canada / Belgrade, Serbia

Schools, Degrees (including field of study), and Years of Graduation
National University of Singapore, Computer Science, 2021;
University of Toronto, Civil Engineering / Sustainable Energy, 2015

Employers, Position / Titles and Dates
Ambient Systems, Founder, 2020-Present
Creative Destruction Lab, Project Manager, 2018-2019;
Various Cleantech Startups, Product management, 2015-2018;
Deloitte Canada, Analyst, 2013-2014

Please tell us about the time you most successfully hacked some (non-computer) system to your advantage.
For my master's project, I wanted to show MEMS sensors can be used to compliment audio signals. The proposed solution has many applications in the localization space, such as smart homes security. The idea was straightforward, but difficult to execute s ince I had only two months to perform the experiments. Since there was no time to ship and assemble my own sensors, I progra mmed my iPhone to collect raw sensor readings various key sensors, such as the accelerometer and magnetometer. The raw dat a resembled a world-class IoT sensing device. This was sufficient to move forward with developing the machine learning model t o test the main hypothesis – all without spending money on expensive equipment.

Please tell us in one or two sentences about the most impressive thing other than this startup that you have built or achieved.
Prior to starting Ambient and my masters degree at the National University of Singapore (NUS), I helped several early-stage star tups launch their products and raise over $1.2M (USD) from various investors, including YC, , and Audi

Are you a technical founder?
Yes

Do you have at least 5% equity in the company you're applying with?
Yes

Will you come to Beijing to participate in the office hours and events hosted by through Nov 2021, if we fund you?
Yes

PROGRESS

what's the company's progress for the past 6 months (in terms of tech development, user acquisition, revenue and funding)
In the last 6 months, we have launched the MVP, filed a patent, brought in 4 more team members, signed 2 MOUs, completed a pilot project with customer, and raised $100k from National University of Singapore (NUS).

How long have each of you been working on this? How much of that has been full-time? Please explain.

About 14 months in total and 6 months full-time. Currently full-time after graduating two weeks ago

Are people using your product?
Yes

当前你的产品研发进展是什么阶段了？是否已经完成样机和测试？是否已经开始销售？预计什么时间可以正式开始销售?
We launched the prototype MVP in May.

如果已经开始销售，已经有多少客户了？增长如何?
We have one active customer and in design stage with two more.

What was your revenue in the last 6 months? (from Dec 2020 to May 2021)
We launched publicly on June 3 so we are still pre-revenue.

Anything else you would like us to know regarding your revenue or growth rate?
We have several interested customers, but they need support with IoT-based solutions in order to acquire data and deploy Ambient. We initially provided the hardware ourselves for the MVP, but now we need a partner to deploy and scale the solution.

If you've applied previously with the same idea, how much progress have you made since the last time you applied? Anything change?
Applying for the first time

IDEA

Which group of people are your customers? What problems do you solve for your customers?
Ambient builds artificial intelligence solutions for smart buildings. Our launch market is climate optimization for indoor farms. Our pro duct helps the operator not only reduce energy consumption but also create an optimal environment to enhance crop growth. Custo mers in the indoor farming market can expect energy savings of 25% to 35%, improved crop yields by 25% to 75%, and 25% reductio n in man-hours. Other segments can expect similar benefits specific to their own desired goals, whether optimal productivity or cost s avings.

Why did you pick this idea to work on? Do you have domain expertise in this area? How do you know people need what you're making?
Buildings are a cornerstone of modern society. However, they are not being optimally used, leading to energy wastage, carbon emissi ons, and overall poor utilization. Existing solutions are either too complex, expensive, or only help the user reduce energy bills, which is a good start but not enough. I believe AI will proliferate across the built environment in the upcoming decade, helping to overcome these challenges. I chose to work on this challenge because I can see the potential for AI to transform how we interact with the built e nvironment. My background in civil engineering provides a foundational understanding, covering building science and energy systems . Through my studies and colleagues, I noticed that there is a big opportunity to apply latest computing innovations in the built sector . This was validated further when I started working on Ambient and speaking with customers across various segments.

What's new about what you're making? What substitutes do people resort to because it doesn't exist yet (or they don't know about it)?
At the moment, most building management systems (BMS) are closed off and don't talk with operational systems, such as ones used f or indoor farming. This creates an information gap between desired outcomes (e.g., crop growth) and equipment settings. Similar ga p can be seen in other industries, such as healthcare, where the building systems are completely disconnected from patient wellbeing or preventing hospital associated infections (HAI). Most customers simply don't have the in-house expertise to integrate and manage these systems. At best, they might adopt a latest Building Automation System (BAS) by companies such as Siemens, Honeywell, or . This is a good first step, but their offerings fall short when it comes to optimization, leaving the customer to scramble to find highly-trained (and expensive) engineers to manage their facilities.

Who are your competitors, and who might become competitors? Who do you fear most?
There are about 10-20 vendors providing AI solutions for buildings. Most of them provide it as an add-on feature as part of their existi ng products, such as building management systems (BMS), lighting systems, or sensors. They are also fairly new in the space, with

rket leaders only launching as recently as 2018, focusing mostly on energy savings.

What do you understand about your business that other companies in it just don't get?
Ambient understands that energy consumption is just one of the things that users want to optimize. Sometimes this is in direct compe tition with other goals, such as indoor air quality or creating an optimal indoor environment for specific uses, such as indoor farm crop growth (which often requires an unusual combination of high CO2, low temperature, and medium RH, among other parameters). We provide a solution that balances these competing priorities using latest machine learning techniques.

What's your business model? How much revenue per year can you achieve in 5 years?
B2B subscription model with optional upfront expenses if the user needs hardware (e.g., sensors, controllers) or system integration. W e aim to capture about 10% of the global AI for buildings market, resulting in $100M - $140M ARR in five years.

How will you get users? If your idea is the type that faces a chicken-and-egg problem in the sense that it won't be attractive to users till it has a lot of users (e.g. a marketplace, a dating site, an ad network), how will you overcome that?
Our AI models will certainly become better over time, but each user will be able to use our product immediately. The hardware takes half a day to install and starts generating insights immediately.

Why do you think this is the right time for you to start your business?
Building owners worldwide are looking for new ways to create value after the pandemic. At the same time, IoT and computing costs c ontinue to decrease, enabling effective data acquisition and capabilities to deploy AI models.

LEGAL

Have you incorporated, or formed any legal entity (like an LLC) yet?
No

What kind of entity and in what city or country was the entity formed?
Private Limited Company in Singapore (application in progress currently)

Please describe the breakdown of the equity ownership in percentages among the founders, employees and any other stockholders. If there are multiple founders, be sure to give the equity ownership of each founder.
Founder: 100%

List any investments your company has received. Include the name of the investor, the amount invested, the premoney valuation / valuation cap.
Offered (Jun 2021): NUS GRIP, $100k

Who is the owner of the company's core product and technology? Have you registered your intelle
Ivan Damnjanovic (founder)
Yes, IP is registered

Does your company have any pending lawsuit or claim?
No

OTHER

If you had any other ideas you considered applying with, please list them. One may be something we've been waiting for. Often when we fund people it's to do something they list here and not in the main application.

Please tell us something surprising or amusing that one of you has discovered.

Please name an amazing high-potential founder you know, explain why you think he/she is amazing and list the contact information

How did you hear about MiraclePlus?
Academic institutions (professors, labs, research institutions, etc.)

What convinced you to apply to MiraclePlus? Did someone encourage you to apply?
NUS GRIP info session

If applicable, please fill in your recommender's name. Recommender(s) may include MiraclePlus team members, , successful entrepreneurs, other investors, etc.
NUS GRIP, Lawrence,

追加消息